

Smart Car Project Report

Student: Li Xiaoming Date: 2026-06

Project Goal:

Build a line-following smart car with obstacle avoidance.

Hardware & Software:

Controller board, two motors, ultrasonic sensor, Python control logic.

Key Features:

- Auto line following with PID control
- Obstacle detection and emergency stop
- Manual remote control mode

Testing Result:

Completed 5 test tracks; average lap time 42 seconds.

Future Plan:

Add camera-based visual navigation.